MGTECON 603 - Problem Set 5

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0 Summary Statistics

The summary statistics of the data are shown in table A.1 in the appendix. Also, the scatterplot and the correlation matrix of the data are shown in figures A.1 and A.2 in the appendix.

1 Problem 1

1.1 Regression of re78 on re75

The regression result of re78 on re75 is shown in table 1, which is visualized in figure 1.

The high R-squared values suggest that re75 is a good predictor of re78 with the regression coefficient being 0.6955(p-value: < 0.001).

	Model 1
(Intercept)	5352.7055^{***}
	(100.8683)
re75	0.6955^{***}
	(0.0060)
Num. obs.	15992
R^2 (full model)	0.4466
R^2 (proj model)	
Adj. R^2 (full model)	0.4466
Adj. R^2 (proj model)	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Table 1: Regression of re78 on re75

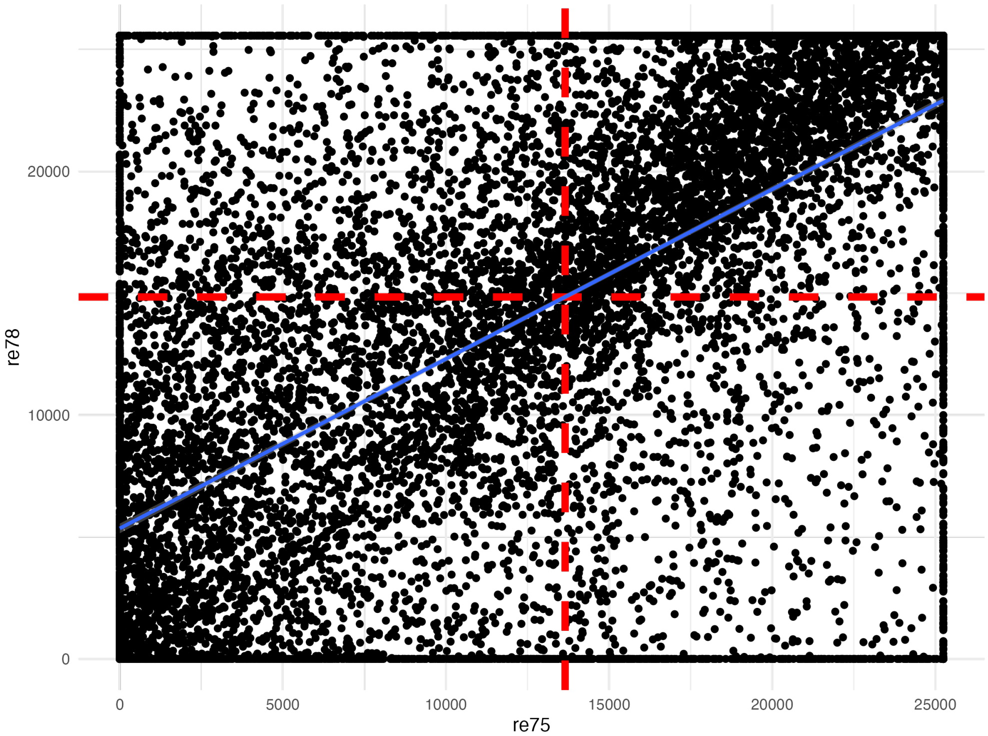

Figure 1: Scatterplot of re75 and re78

Notes: The plot shows the scatterplot of re75 and re78. The line is the regression line. Also note that these earnings have a clear maximum and minimum cutoffs, which make the plot display a strict rectangle form. This is potentially due to winsorization done on the data. The red dashed lines indicate the mean of re75 and re78.

1.2 Leverage of re75

The leverage equation for single covariate case is given by:

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2} \tag{1}$$

where i is the index of observation, x_i is the value of the covariate, and $\overline{x}$ is the mean of the covariate. This suggests that the end points of the data are the observations with the highest leverage. Since the mean is closer to the right end point of re75, it is likely that the observations at the left end(i.e., the ones with the lowest re75) are the ones with the highest leverage. The table 2 shows the top 5 observations with the highest leverage, which are consistent with the above observation – the observations with the lowest re75(i.e., zero) have the highest leverage.

This can be also seen in figure 2, which shows the scatterplot of re75 and its leverage. The plot is U-shaped with the lowest re75 having the highest leverage, which is consistent with the above observation.

	re75	leverage_re75
1	0.000000	0.000198
2	0.000000	0.000198
3	0.000000	0.000198
4	0.000000	0.000198
5	0.000000	0.000198

Table 2: Top 5 observations with the highest leverage(single regressor)

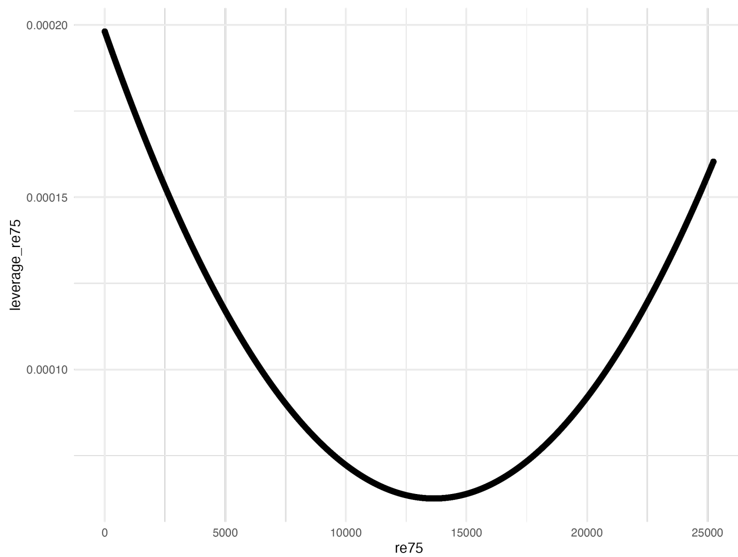

Figure 2: Scatterplot of re75 and its leverage

2 Problem 2

2.1 Regression of re78 on re75, re74, education, black, hispanic, and age

The regression result of re78 on re75, re74, education, black, hispanic, and age is shown in table 3.

The regression results indicate that past earnings (re75 and re74) and education level are significant positive predictors of earnings in 1978. Conversely, being Black and older age are associated with significantly lower earnings, while being Hispanic does not have a statistically significant effect.

2.2 Leverage of re75, re74, education, black, hispanic, and age

The leverage equation for multiple covariates case is given by:

$$[\mathbf{H}]_{ii} = \left[\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right]_{ii} \tag{2}$$

	Model 1
(Intercept)	6461.5815^{***}
	(306.6338)
re75	0.4702^{***}
	(0.0153)
re74	0.2913^{***}
	(0.0150)
education	113.9149^{***}
	(20.2172)
black	-826.2451^{***}
	(198.3461)
hispanic	-209.0108
	(218.7439)
age	-102.6443^{***}
	(5.3389)
Num. obs.	15992
R^2 (full model)	0.4752
R^2 (proj model)	
Adj. R^2 (full model)	0.4750
Adj. R^2 (proj model)	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Table 3: Regression of re78 on re75, re74, education, black, hispanic, and age

where i is the index of observation, x_i is the value of the covariate, and $\overline{x}$ is the mean of the covariate. This suggests that the end points of the data are the observations with the highest leverage. Since the mean is closer to the right end point of re75, it is likely that the observations at the left end(i.e., the ones with the lowest re75) are the ones with the highest leverage. The table 4 shows the top 5 observations with the highest leverage, which are consistent with the above observation – the observations with the lowest re75(i.e., zero) have the highest leverage.

The five observations with the highest leverage are shown in table 4. The last column(high_leverage) indicates whether the observation is a high leverage observation, defined as leverage value greater than $3K/n$, where K is the number of covariates and n is the number of observations. Although the cutoff is quite arbitrary, the five observations with the highest leverage are all high leverage observations.

Is this concerning? I do think so according to the cutoff. **What concerns me more** is that the data seems to have been winsorized, which is potentially due to the outliers. However, given that the winsorized numbers would all fall into the group of high leverage observations, I believe that this implies that the winsorization cutoff is what drives the regression results.

	re75	re74	leverage_multivariate	high_leverage
1	25243.5500	0.0000	0.0030	TRUE
2	25243.5500	0.0000	0.0027	TRUE
3	1090.3060	25862.3200	0.0026	TRUE
4	25243.5500	0.0000	0.0026	TRUE
5	25241.7600	0.0000	0.0026	TRUE

Table 4: Top 5 observations with the highest leverage(multiple regressors)

3 Problem 3

3.1 Residuals and their Standardization

Figure 3 shows the histogram of the standardized residuals.

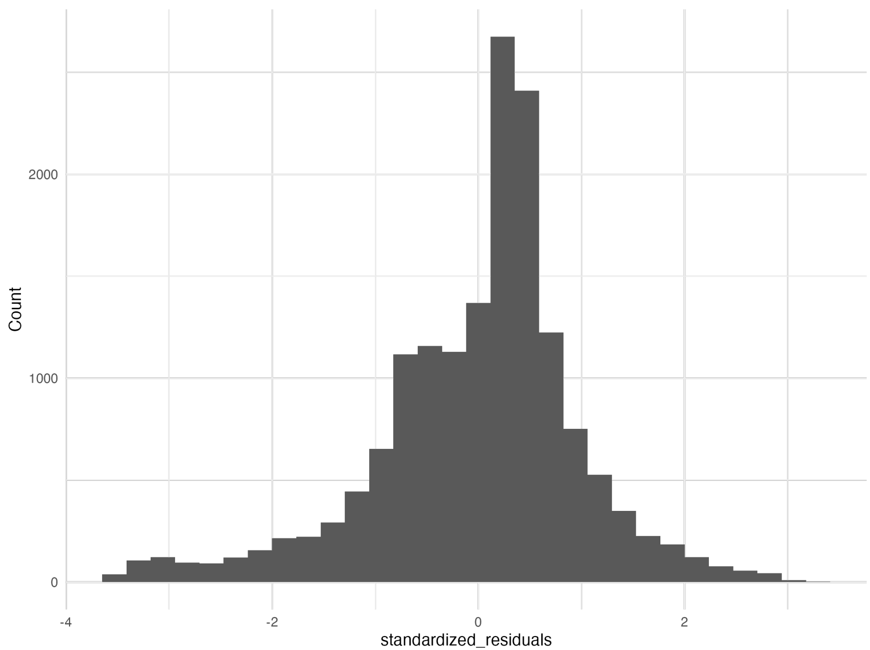

Figure 3: Histogram of the standardized residuals

The highest five values in absolute value are shown in table 5. Here, we do not have any observations from the top 5 observations with the highest leverage. This is likely due to the fact that high leverage observations generally have lower conditional variance of the residuals[1], and thus the residuals with higher absolute value are less likely to be high leverage observations.

[1]conditional on the covariates

	id	standardized_residuals	leverage_multivariate	high_leverage
1	6356	3.2574	0.0006	FALSE
2	9576	3.2400	0.0012	FALSE
3	7386	3.2375	0.0007	FALSE
4	12916	3.1721	0.0004	FALSE
5	10718	3.1692	0.0004	FALSE

Table 5: Top 5 observations with the highest standardized residuals

4 Problem 4

4.1 10-fold Cross-Validation

For problems 4 and 5, I use 10 fold cross-validation to estimate the RMSE of the model. In detail, the overall out-of-sample RMSE is estimated as follows:

1. **Partition the Data**: The entire dataset is randomly shuffled and divided into 10 equally sized, non-overlapping subsets, often called "folds". Let these folds be denoted as $F_1, F_2, \ldots, F_{10}$. Given a specific fold F_k, denote the remaining 9 folds as F_{-k}.

2. **Iterative Training and Testing**: A loop is executed 10 times. In each iteration k (from 1 to 10):

   - Fit the model on the training set F_{-k}:

   $$\arg\min_{\beta^{(-k)}} \sum_{i \in F_{-k}} (y_i - \beta^{(-k)\top} x_i)^2$$

   - The trained model is then used to predict the outcomes for the observations in the held-out fold, F_k:
   $$\hat{y}_i = \beta^{(-k)\top} x_i$$

3. **Compute Overall RMSE**: Finally, the out-of-sample RMSE is calculated by aggregating the prediction errors from all 10 folds.

The formula is:

$$RMSE_{CV} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{3}$$

where y_i is the true outcome for observation i, and $\hat{y}_i$ is the out-of-sample prediction for observation i that was generated when its fold was used as the test set.

4.2 Simple Regression and OOS RMSE

The RMSE that I obtained from the simple regression is 7177.183. This result is derived by the following code:

```
k_folds <- 10
data$fold <- sample(1:k_folds, n, replace = TRUE)
all_predictions_p4 <- data.table()

for (k in 1:k_folds) {
    train_data <- data[fold != k]
    test_data <- data[fold == k]

    ## Fit the model
    l <- fixest::feols(fml = as.formula(paste0("re78 ~ re75")),
                        data = train_data,
                        vcov = "hetero")

    predictions <- test_data[, predicted_re78 := predict(l, newdata = test_data)]

    temp_data <- data.table(id = test_data$id,
                            re78 = test_data$re78,
                            predicted_re78 = predictions)
    all_predictions_p4 <- rbind(all_predictions_p4, temp_data)
    }

all_predictions_p4[, error := re78 - predicted_re78]
RMSE_p4 <- sqrt(mean(all_predictions_p4$error^2))
```

The scatter plot of predictions and errors is shown in figure 4.

5 Problem 5

5.1 Multiple Regression and OOS RMSE

The RMSE that I obtained from the multiple regression is 6691.683. This result is derived by the following code:

```
all_predictions_p5 <- data.table()

for (k in 1:k_folds) {
    train_data <- data[fold != k]
    test_data <- data[fold == k]

    rhs <- "re75 + re74 + education + black + hispanic + age"

    ## Fit the model
    l <- fixest::feols(fml = as.formula(paste0("re78 ~ ", rhs)),
                        data = train_data,
                        vcov = "hetero")
```

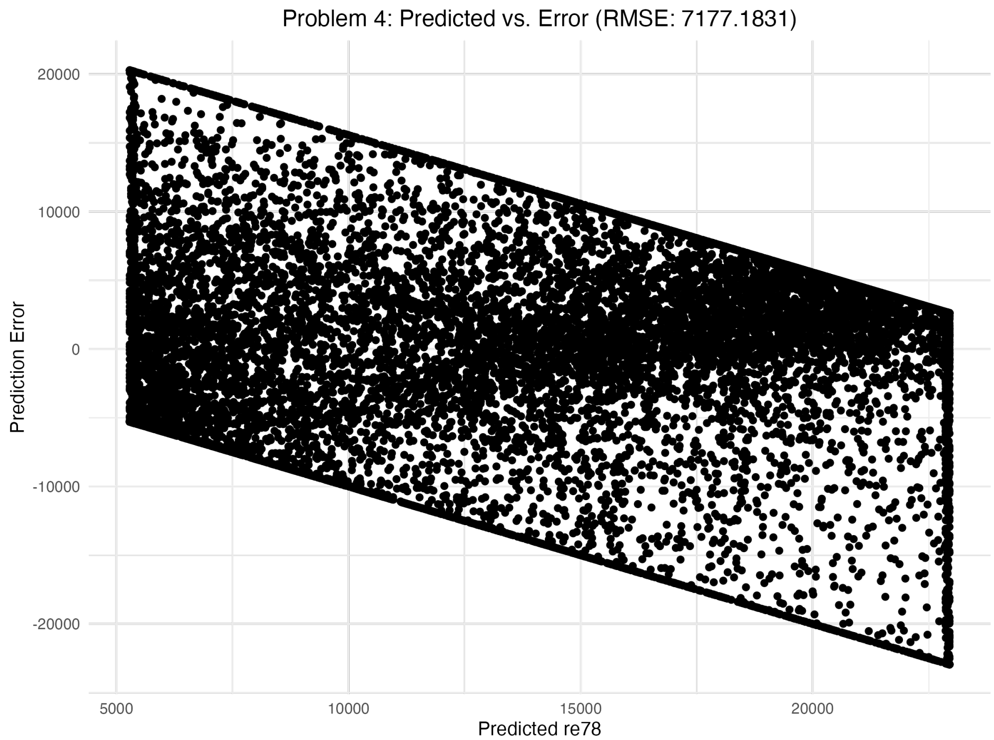

Figure 4: Scatterplot of predicted re78 and error

```
predictions <- predict(l, newdata = test_data)

temp_data <- data.table(id = test_data$id,
                        re78 = test_data$re78,
                        predicted_re78 = predictions)
all_predictions_p5 <- rbind(all_predictions_p5, temp_data)
}

all_predictions_p5[, error := re78 - predicted_re78]
RMSE_p5 <- sqrt(mean(all_predictions_p5$error^2))
```

The scatter plot of predictions and errors is shown in figure 5.

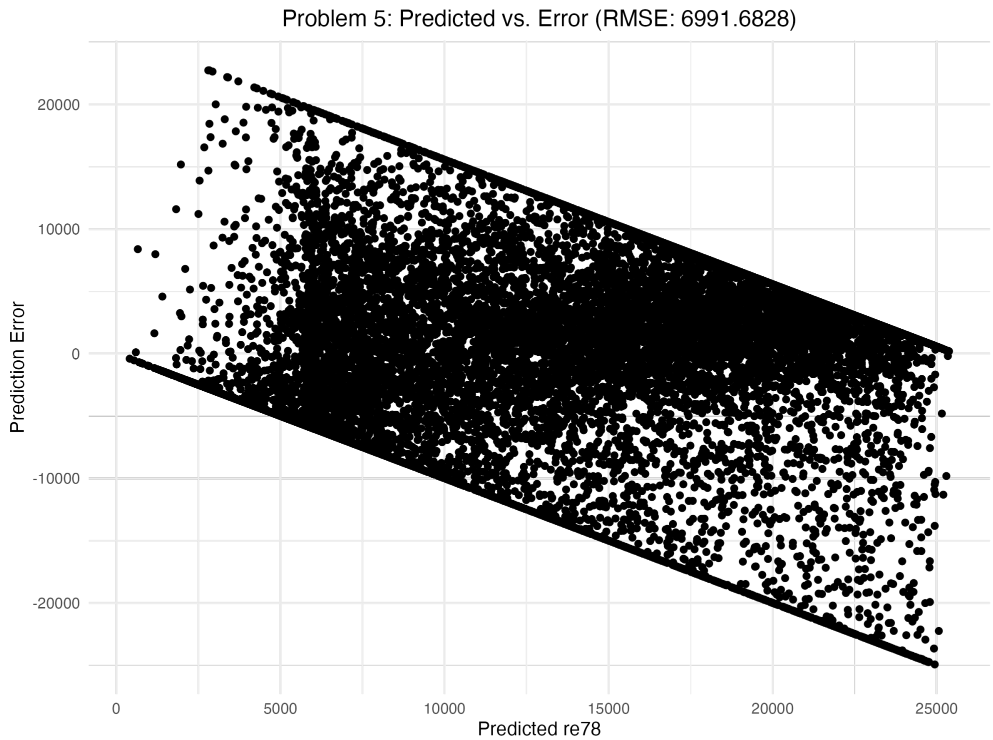

Figure 5: Scatterplot of predicted re78 and error

A Summary Statistics

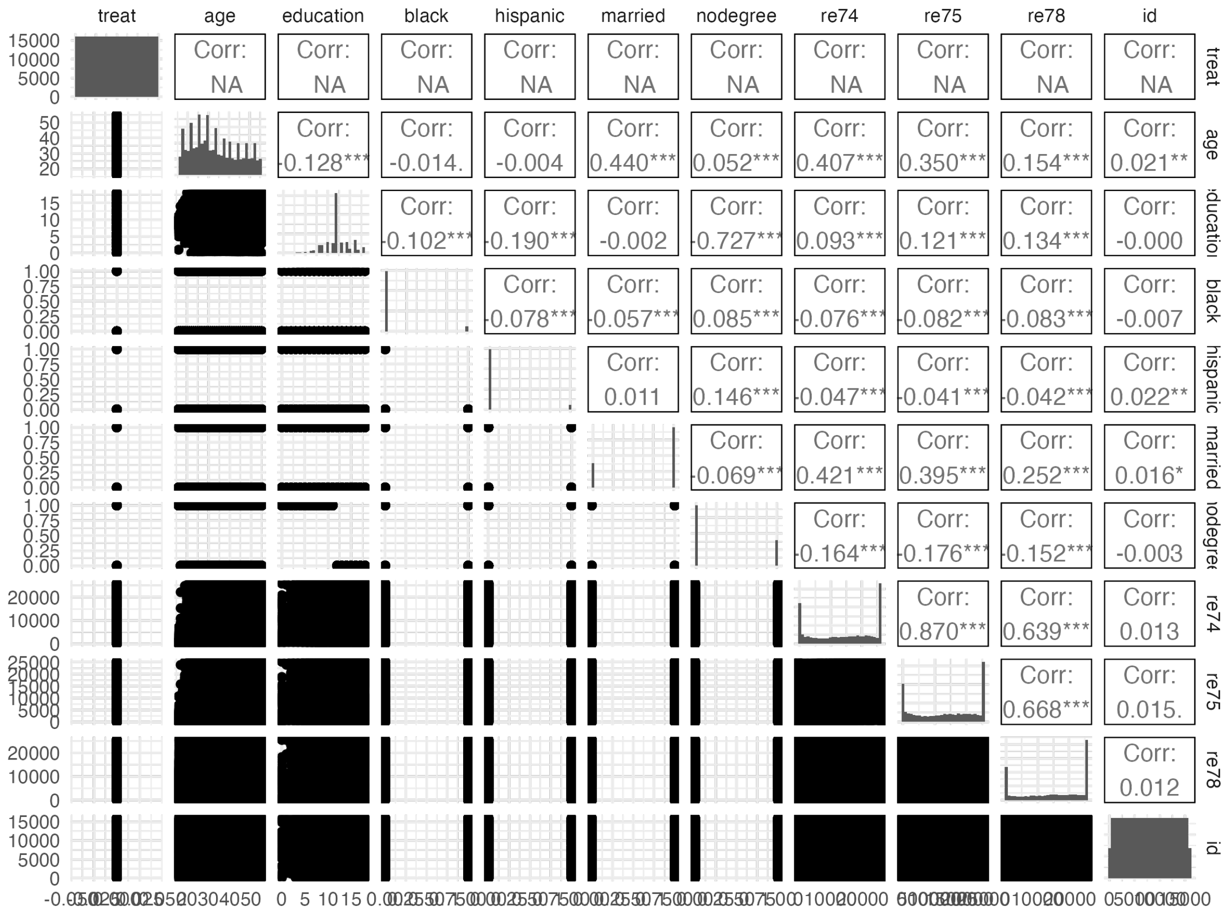

Figure A.1: Scatterplot Matrix

variable	numNA	numZeros	fracZeros	mean	sd	min	max	10%	20%	30%	40%	50%	90%	95%	99%	99.9%
treat	0	15,992	1.0000	0.0000	0.0000	0	0.00	0.0	0.000	0.000	0.00	0.00	0.00	0.00	0.00	0.00
age	0	0	0.0000	33.2252	11.0452	16	55.00	19.0	23.000	26.000	28.00	31.00	50.00	53.00	55.00	55.00
education	0	36	0.0023	12.0275	2.8708	0	18.00	8.0	10.000	12.000	12.00	12.00	16.00	17.00	18.00	18.00
black	0	14,816	0.9265	0.0735	0.2610	0	1.00	0.0	0.000	0.000	0.00	0.00	0.00	1.00	1.00	1.00
hispanic	0	14,840	0.9280	0.0720	0.2586	0	1.00	0.0	0.000	0.000	0.00	0.00	0.00	1.00	1.00	1.00
married	0	4,610	0.2883	0.7117	0.4530	0	1.00	0.0	0.000	1.000	1.00	1.00	1.00	1.00	1.00	1.00
nodegree	0	11,261	0.7042	0.2958	0.4564	0	1.00	0.0	0.000	0.000	0.00	0.00	1.00	1.00	1.00	1.00
re74	0	1,913	0.1196	14,016.8003	9,569.7959	0	25,862.32	0.0	2,411.857	6,716.955	11,207.01	15,123.58	25,862.32	25,862.32	25,862.32	25,862.32
re75	0	1,748	0.1093	13,650.8034	9,270.4032	0	25,243.55	0.0	2,544.406	6,565.650	10,994.37	14,557.11	25,243.55	25,243.55	25,243.55	25,243.55
re78	0	2,172	0.1358	14,846.6597	9,647.3915	0	25,564.67	0.0	2,915.850	8,198.427	12,814.25	16,421.97	25,564.67	25,564.67	25,564.67	25,564.67
id	0	0	0.0000	7,996.5000	4,616.6371	1	15,992.00	1,600.1	3,199.200	4,798.300	6,397.40	7,996.50	14,392.90	15,192.45	15,832.09	15,976.01

Table A.1: Summary Statistics

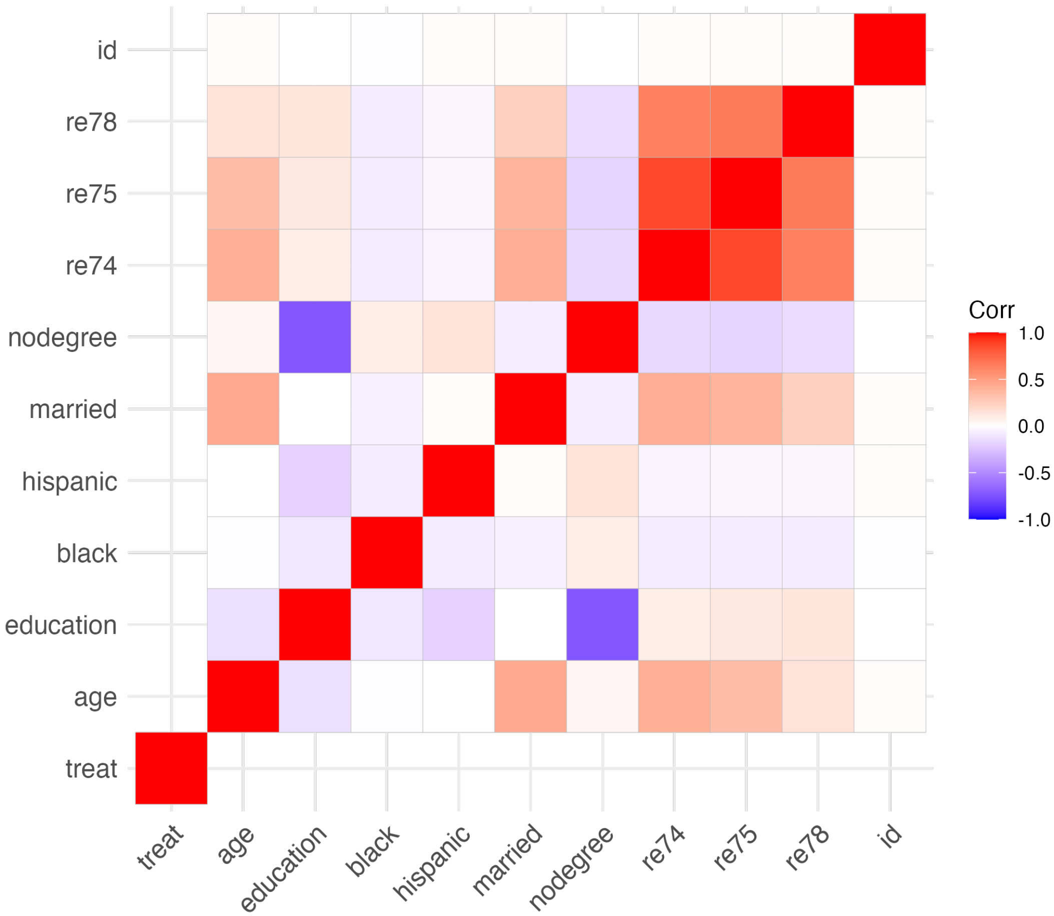

Figure A.2: Correlation Matrix Plot